

**Root Duality: Unifying Indexical Identity and Computational Anchoring —
A Focus on the Mechanisms of Complex Word Formation**

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Abstract:

This paper analyzes the study of root properties in constructional morphology, examining the multilingual process of complex word construction. Building on Belder & Craenenbroeck (2015), which characterizes the essence of roots as the “empty position” created by initial Merge-driven displacement, the paper presents a systematic analysis and argues that roots exhibit a dual nature as both indexical units and computational anchors.

Key words: root, empty position, index, Computational Anchoring

1. Research background and questions

Regarding roots, the constructivist tradition roughly divides into three positions:

(1) Represented by Harley (2014) and others, roots are indices. In syntax, a ROOT is represented by an arbitrary number or index to distinguish each root from others. A word’s root may have different phonological forms in different syntactic environments. Because the semantic content of roots is highly variable and may resemble a “semantic replacement” phenomenon, relying on phonology or semantics alone is insufficient to identify the root initially. Therefore, Harley proposes treating roots as purely structural units in syntax for computation, with neither semantic nor phonological features. Belder & Craenenbroeck (2015) begin from principled merge and characterize roots as having four properties: (i) roots lack grammatical features, (ii) roots lack syntactic categories, (iii) roots are defined by structure rather than lexicon,

and (iv) roots are governed by functional categories (not the other way around).

(2) Marantz (1997); Embick (2004) consider roots to have intrinsic ontology: Marantz divides roots into entity roots (e.g., objects), state roots (e.g., properties), and event roots (e.g., actions); Hovav & Levin (2010) propose roots can be classified as “manner” and “result,” whose semantic types determine the syntactic behavior of verbs.

(3) Acedo-Matellán & Mateu (2015); Borer (2013) argue that the semantic properties of roots arise from syntactic structure rather than intrinsic semantics. Roots are category-free abstract units; semantic interpretation entirely depends on their combination position with functional morphemes (such as *v*, *n*).

The above synopsis outlines Harley’s view of indexation that is semantically empty and phonologically void, and the baselines on semantics and phonology. Marantz (2024)¹ argues that in syntax, “indices” are not the elements used by syntactic operations; syntactic manipulations do not identify numbers like #53 or #67. Roots should be grounded in a baseline phonological form, i.e., the ontological view introduced above is foundational.

Regarding the view of meaninglessness, Borer (2013), in the Exo-Skeletal model, points out that roots have no intrinsic meaning; root semantics are retrieved at some stage of syntactic derivation by consulting encyclopedic knowledge.

In sum, the linguistic literature holds divergent views on ROOT, differing mainly in emphasis. Concerning root properties, I align with Belder & Craenenbroeck (2015) in their claim that “We show that as a by-product of this asymmetry, the first Merge operation in each cyclic domain—called Primary Merge here—creates a radically empty structural position at the foot of the structure in which a root can be inserted at the level of Vocabulary Insertion.” I further argue that

¹ The perspective in this section derives from correspondence between Alec Marantz and the author.

the function of ROOT in language generation should be realized in staged assignments of distinct roles: after the end of the syntactic operation stage, during the Sensorial-Motor Interface (S-M) and the Conceptual-Intentional Interface (C-I) identification processes, the root plays a dual role as both an indexical unit and a computational anchor.

Building on Belder & Craenenbroeck (2015), this paper reexamines root properties, proposes the dual-nature view of roots, and argues for a mechanism of word formation in complex words under this duality. The remainder of the paper is organized as follows: Section 2 presents the root duality view; Section 3 discusses the roles of the root's indexical unit and computational anchor in the word-formation process of complex words; Section 4 briefly discusses the implications of root duality for studies of syntactic construction; Section 5 concludes.

2. What Is a Root?

Root properties and the minimalist-DM perspective

What is a root? I regard this as a prerequisite for discussing root properties. Belder & Craenenbroeck (2015) argue that, within the Minimalist Program and Distributed Morphology (DM), a root is defined by syntactic structure rather than by the lexicon. They propose an asymmetric Primary Merge view and show that during the syntactic building process, Primary Merge is directional: it always merges a set of grammatical features (for example, the feature set $\{\alpha\}$) with an empty set ($\emptyset$) to form the ordered pair $\langle\{\alpha\}, \emptyset\rangle$, with $\{\alpha\}$ projecting and bearing a label. The empty position created by the initial merge is later filled by a root, thus they treat the root as an “empty slot” in the syntactic structure, devoid of inherent grammatical information and categories, with categorization supplied by functional morphemes. They also discuss and demonstrate four properties: (1) roots lack grammatical features, (2) roots lack syntactic categories, (3) roots are defined by structure rather than lexicon, and (4) roots are governed by functional categories (not the reverse).

I endorse Belder & Craenenbroeck (2015) and propose that roots (ROOT) are interface-identification indices that are also staged. In the syntactic operation phase (when the syntactic terminal inventory in List 1 governs operations), their nature

should be an empty position, as Belder & Craenenbroeck (2015) describe the “We show that as a by-product of this asymmetry, the first Merge operation in each cyclic domain¹—called Primary Merge here²—creates a radically empty structural position at the foot of the structure in which a root can be inserted at the level of Vocabulary Insertion.” radical empty structural position located at the bottom of the structure. In the morphology/spelling-out phase (Vocabulary Insertion), the root can be inserted into this position. I further argue that after synchronous merging and uniform operations, this empty position provides an interface index for late insertion, functioning as an index unit. This index unit finishes linearization during the morphological operation phase (where List 2 supplies the word-item inventory for morphology). At this stage, the root serves as the basis for inflectional morphophonology, i.e., the root’s phonological form provides the morphophonological interface basis for alternations, albeit with the phonological form being a late fill-in². Likewise, during the encyclopedic Knowledge Base phase (where List 3 provides encyclopedic knowledge), the root also completes a semantic operation baseline. See Figure 1.

² Marantz’s idea that the morphophonological adjustment from “go” to “wen(t)” begins with the index, so the underlying form of the root has a phonological realization, with the phonology filled in at a late stage.

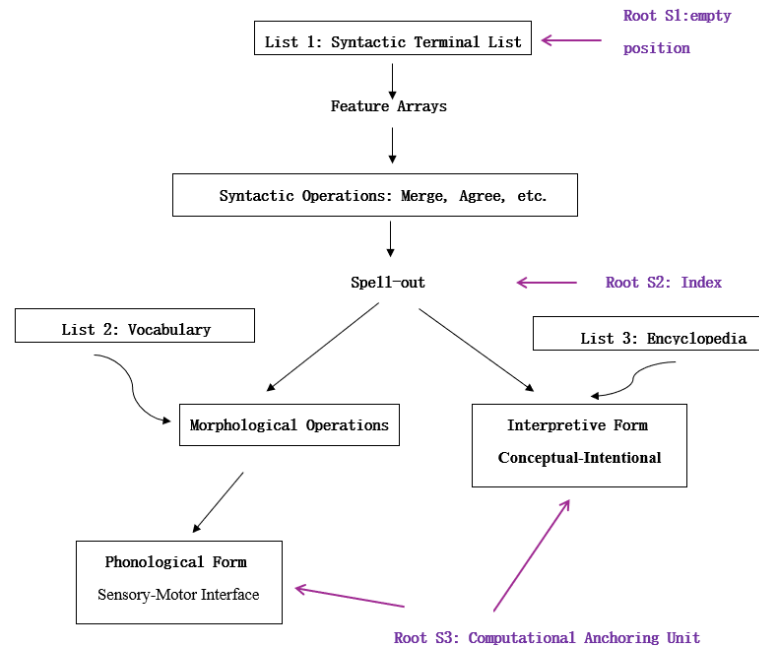


Figure 1 schematically represents this phase-based derivational process.

In sum, I regard ROOT as a staged element that is fundamentally unified as both an indexical unit and a computational anchor.

3. Root indices and Computational Anchoring (index function and baseline)

In the process of syntactic construction (Building), the primary merge stage gives rise to what is called a null derivation, in which no substantive lexical items are introduced at the initial stage of syntactic operation. The structure begins with a merge with the empty set, so this derivation is initially “empty” and lacks phonological or semantic content. That empty position functions as an index, and after word formation it is transferred to the interface to receive phonology and meaning. Following Marantz (2007), we can also treat words as phases; of course, word formation must respect the Phase Impenetrability Condition (PIC). In Building, this “empty position” (which we denote ROOT_c, the initial root) exhibits two properties: operability (OY) and baseline startup capability (BSC). These properties are realized by the structure created by Merge and govern how roots combine with other elements to form words. After the operation concludes, the initial root position remains an interface-late phonology and meaning slot, giving

rise to a baseline startup, becoming the morpho-phonological and semantic computation anchor, and is identifiable at the interface to form indices. Given that Chinese employs numerous compounding strategies, such as in “da huang gou” (大黄狗, big yellow dog) and “huang da yi” (黄大衣, yellow coat) we illustrate word formation with these two complex words.

$$(1) [\# \sqrt{\text{huang}} [\text{nP} < \sqrt{\text{da}} \sqrt{\text{yi}} >, \text{n}]]$$

$$[\# \sqrt{\text{黄}} [\text{nP} < \sqrt{\text{大}} \sqrt{\text{衣}} >, \text{n}]]$$

yellow coat

$$(2) < [\# \sqrt{\text{da}} [\sqrt{\text{haung}} \sqrt{\text{gou}}]], \text{n} >$$

$$< [\# \sqrt{\text{大}} [\sqrt{\text{黄}} \sqrt{\text{狗}}]], \text{n} >$$

big yellow dog

(1) huang + dayi (“yellow coat”) is an outer-domain compound, where the co-component for coat is directly transferred from the yellow domain. (2) da huang gou (“big yellow dog”) is an inner-domain compound, so big and yellow attach within the root-domain. The coat, under the Phase Impenetrability Condition (PIC), cannot be operated on further.

In the first phase of roots (Phase S1 as in Figure 1), the roots “huang(yellow), da (big), yi(coat)” and “da(big), huang(yellow), gou(dog)” are merged at List 1. We regard this as an initial merge between an initial root (ROOTc) and a category-less morpheme n. According to the two features operability (OY) and baseline startup capability (BSC), the following changes occur:

(1a)

Primary Merge: $[\{\emptyset, \emptyset\}, \text{n}] \longrightarrow <\{\emptyset, \emptyset\}, \text{n}>$ **【OY】**

Merge-and-sequential-labeling stage: $\{\{\emptyset, \emptyset\}, \text{n}\} \longrightarrow <\{\emptyset_1, \emptyset_2\}, \text{n}>$

Structure Building: $[\{\emptyset\}, <\{\emptyset_1, \emptyset_2\}, \text{n}>] \longrightarrow \{\{\emptyset\}, <\{\emptyset_1, \emptyset_2\}, \text{n}>\}$

Derivation by Phase: $\{\{\emptyset\}, <\{\emptyset_1, \emptyset_2\}, \text{n}>\} \longrightarrow \{\{\emptyset\}, <\{\emptyset_1, \emptyset_2, \text{n}>\} \}$ **【BSC】**

At this stage, transfer to the interface the word-formation segments $\{\emptyset_1, \emptyset_2\}$ for interface operations (Phase S2 and S3 in Figure 1).

Merge-and-sequential-labeling stage: $\{\{\emptyset\}, <\{\emptyset_1, \emptyset_2\}, \text{n}>\} \longrightarrow \{\{\emptyset_3\}, \text{nP}\}$

In this example, the word-formation segments progress to the top node, transferring the operable empty set $\{\emptyset_3\}$ for interface operations and completing the word formation.

In word-formation cases like “dayi(coat)” the initial merge yields two empty positions that merge with the productive morphemes. I argue this arises because the initial root carries operability features $[+OP]$, whose role is to trigger the merge between the root’s empty position and other constituents. Since $[+OP]$ is a syntactic structural feature and lacks a category, it cannot bear a label; thus, when merging with a productive morpheme, the morpheme’s category is shared and labeled accordingly. Of course, when merging with another initial root carrying $[+OP]$, the resulting empty position (which I call the Operability Empty Set) contains two constituents (the two parts of “coat” correspond to two empty positions as in 1b)³. In syntactic operations, there is effectively only one empty position (the Operability Empty Set); phonological realization occurs at the interface after the split and filling. We treat the “empty position” in word-formation as the Operability Empty Set; the number of elements within this set is not tied to the count of empty positions available in the workspace, but serves as a marker of the root’s index properties. This empty structure must be constructed through merging, which also satisfies the principle that “In every derivational workspace, there is exactly one root, and for every root there is exactly one derivational workspace. (One Derivational Workspace, One Root(ODWOR)Belder & Craenenbroeck 2015)”.

$$(1b) \{\emptyset_1\}, \{\emptyset_2\} \longrightarrow \{\emptyset_1, \emptyset_2\} = \text{OperNS}$$

$$[\text{OperNS}, n] \longrightarrow \langle \{\emptyset_1, \emptyset_2\}, n \rangle$$

It should be noted that the subscripts 1, 2, 3 on empty positions simply reflect the order of merging, not the identification index. Accordingly, I designate this stage as the Merge-and-Sequential-Labeling stage.

³ It should be noted that the OperNS (Operability Null Set) introduced here is indeed an empty position in the syntactic structure. We define this set precisely to reflect the index-unit function of the initial root.

In the stage $\{\emptyset_3, nP\}$, after $\emptyset_3$ merges with nP , it triggers the transfer to the interface in the syntactic construction process toward the S-M (sensory-motor) interface. During the path to the **Sensory-Motor** Interface, the index of the initial root and the features of the category morpheme undergo a baseline-startup activation (BSC) that prompts searching for matching vocabulary items (VIs) in List 2, i.e., in its own lexicon, for insertion. At the same time, the concept–intention system provides meaning through encyclopedic knowledge.

Word formation in “da huanggou(Big Yellow Dog)” follows the same OperNS and category-morpheme merging, with the difference that there are three initial roots. First, two of them merge to form an OperNS1, then OperNS1 merges with the third initial root to form OperNS2, and finally OperNS2 merges with a category morpheme (as shown in 2a).

$$\begin{aligned}
 (2a) \quad & \{\emptyset_1\}, \{\emptyset_2\} \longrightarrow \{\emptyset_1, \emptyset_2\} = \text{OperNS}_1 \\
 & \{\emptyset_3\}, \text{OperNS}_1 \longrightarrow \{\emptyset_3, \text{OperNS}_1\} = \{\emptyset_3, \{\emptyset_1, \emptyset_2\}\} = \text{OperNS}_2 \\
 & [\text{OperNS}_2, n] \longrightarrow \langle \{\emptyset_3, \{\emptyset_1, \emptyset_2\}\}, n \rangle
 \end{aligned}$$

In Derivation Workspace: OperNS merges with category morphemes, and each OperNS can itself be treated as a derived workspace. Thus, the empty positions $\emptyset$ presented above can be regarded as arising from merges with $[+OP]$, i.e., all $\emptyset$ correspond to $\langle [+OP], \emptyset \rangle$. This aligns with the ODWOR framework. It should also be noted that the merge order for “ $\sqrt{\text{da}}(\sqrt{\text{big}})$ ”, “ $\sqrt{\text{huang}}(\sqrt{\text{yellow}})$ ”, and “ $\sqrt{\text{gou}}(\sqrt{\text{dog}})$ ” is established at the interface through encyclopedic knowledge. In the construction, because merges are pairwise, an OperNS will appear within the OperNS itself during the syntactic process (these remain empty positions by structural necessity and are not labeled yet). Since labeling cannot occur at this stage, the final labeling and meaning assignment are activated only when the phrase transfers to the interface, at which point baseline startup capability (BSC) enables phonology and semantics to be supplied.

Some scholars (Zhang 2007; Bauke 2014; De Belder & Craenenbroeck 2015) adopt a view in which roots can merge directly in syntax. Others oppose this approach. Nobrega (2015) argues roots are flawed as independent syntactic units

and cannot operate syntactically on their own. To address this, Nobrega (2015) proposes edge features as a remedy, suggesting edge features force roots to merge with a determiner (category) morpheme to be pronounceable and usable in syntax. Chenchen Julio Song (2020) introduces a functional center Cat; all roots first merge with Cat before proceeding with operations.

As noted above, in the syntax-derivation stage initial Merge is always asymmetric: it merges an abstract morpheme with an empty set, and regardless of how many initial roots are involved, there is a single empty position (an operable empty set) at the syntactic stage. These empty positions are filled at the interface through phonology and interpretation. The operable empty set internal to the initial roots functions as an interface index that appears during spell-out, which is where the [+OP] feature, which we propose as a modification of the Cat property in Song (2020), plays a role. Our argument is that creating operability for roots is a property of the structure itself ([+OP]); under Null Derivation, these initial roots are, in principle, empty positions that do not require adding abstract morphemes to achieve asymmetric merging. The merging of several empty sets remains an empty set; there is no need to posit extra abstract morphemes to complete the syntactic operation. These initial roots ultimately become indices at spell-out and are interpreted at the interface, leaving the syntactic structure themselves without the need to assign additional functional roles to resolve interface visibility.

The Miracle Creed in Chomsky (2024) states: “We can entertain the doubts: the only numbers are 1, 2, and infinity. The only operations and relations that accord with SMT are unary, binary, and unbounded. Unary operations do nothing in I-language,¹² so we can omit them. Binary relations permit the fewest options for carrying a derivation forward, a sufficient reason for keeping to $n=2$ for admissible relations and for the operations that create them, unless shown otherwise.” and “If 1-language is basically a thought-generating system, it optimally should observe the principle T:[T] All relations and structure-building operations (SBO) are thought-related, with semantic properties interpreted at Cl.” The first part provides a theoretical basis for the Null Derivation perspective, and the second part offers a

theoretical basis for our analysis of compounds and roots.

We can refine the Building stage in (1a):

Primary Merge: $[\text{OperNS}, n] \longrightarrow \langle \text{OperNS}, n \rangle$

Structure Building: $[\{\emptyset\}, \langle \text{OperNS}, n \rangle] \longrightarrow \{\{\emptyset\}, \langle \text{OperNS}, n \rangle\}$

Derivation by Phase: $\{\{\emptyset\}, \langle \text{OperNS}, n \rangle\} \longrightarrow \{\{\emptyset\}, \langle \text{OperNS}, n \rangle\}$

Transfer the OperNS to the spell-out phase (Spell-out) for index labeling and merge-and-sequential-labeling.

Merge-and-sequential-labeling stage: $\text{OperNS} \longrightarrow \langle \{\emptyset_1, \emptyset_2\}, n \rangle$

The empty positions $\{\emptyset_1, \emptyset_2\}$ are transferred directly to the interface for phonological realization and semantic interpretation.

$\{\{\emptyset\}, \langle \{\emptyset_1, \emptyset_2\}, n \rangle\} \longrightarrow \{\{\emptyset_3\}, nP\}$

Derivation by Phase: $\{\{\emptyset_3\}, nP\}$

The word-formation segment derives to the top node and transfers the operable word-formation segment $\{\emptyset_3\}$ to the interface for operations, completing the word formation.

3.1: Re-examining the root properties of Chinese compounds

Chen & Liu (2020) provide a detailed discussion of the formation of four types of Chinese compounds: coordinate compounds, attributive compounds, verb-result compounds, and verb-object compounds. They note that (1) coordinate compounds are formed by merging two initial roots first, then combining with a determiner morpheme; (2) attributive compounds involve a modifier occupying a demonstrative position, followed by merger with the category-mapped component (the initial root merged with the determiner morpheme); (3) verb-result compounds consist of the merger of two category-mapped verbs (V1 and V2), where the determiner morpheme of V2 denotes a light event; (4) verb-object compounds are built after the verb phrase is lexicalized and then combined with a determiner morpheme. Aside from verb-result and verb-object compounds being constructed through merges of words or phrases, both coordinate and attributive compounds in Chinese start from roots in their derivational process. From Chinese coordinate and attributive compounds alone, one obtains that the initial root merges only with a determiner

morpheme that maps onto the functional category, which is consistent with Belder & Craenenbroeck (2015).

Our approach to attributive compounds diverges from Chen & Liu (2020) in how we treat the processing of words like “dayi(coat)” Chen & Liu (2020) propose first forming “√yi(clothing)” merging with the determiner morpheme *n*, sharing the determiner morpheme *n*, and then undergoing a secondary merge with the modifier “√da” requiring a second merger. By contrast, based on the theory of Null Derivation and the ODWOR framework, we argue that for words such as “dayi(coat)” in the derivational stage, the operable empty set merges with the determiner morpheme, where the elements inside the operable empty set are split during spell-out and serve only as interface indices. The internal modifier relations are determined by the semantic interface. The same logic applies to coordinate compounds. Of course, one could also regard each root as undergoing null derivation followed by a Markov-style merger⁴, after which further derivation occurs. The operability feature [+Oper] arises from the index properties of the root and is a property of the merging operation itself rather than a constituent in the syntactic structure, so it cannot bear a label and cannot be transferred to the interface.

We can treat both attributive and coordinate compounds as syntactically identifiable when their grammatical categories are determined at the syntactic node. In such cases, the attributive and coordinate relations are provided by encyclopedic knowledge, and there is no need to assign a dedicated position for a modifying constituent in the structure. This is because, during word-formation derivation, the operable empty set in the word-formation segment will inherit the determiner-morpheme features (regardless of how many elements it contains) and, after labeling, be transferred to the interface.

⁴ Note that Markov-style merging is a multi-dimensional merger in which independent workspaces merge and then project onto a single dimension, aligning with the ODWOR principle.

3.2: Root properties of compounds from a comparative perspective

We illustrate with German compounds to assess whether the proposed root-building rules remain applicable.

(4)

Dachziegel (roof tile) Eisenbahn (train/railway) Geburtstag (birthday)

The three German examples in (4) are all attributive compounds, i.e., Root+XP structures. We focus on “Geburtstag (birthday),” which is formed by merging “√Geburt” with “√Tag” and the determiner morpheme “n” and then an linking element [+connect]⁵ (see (4a)).

Primary Merge: $[\emptyset, n] \longrightarrow \langle \emptyset, n \rangle$ 【OY】

Merge-and-sequential-labeling stage: $\langle \emptyset, n \rangle \longrightarrow \langle \emptyset_1, n \rangle$

Structure Building: $[[[+connect]], \langle \emptyset_1, n \rangle] \longrightarrow \{ \{ [+connect] \}, \langle \emptyset_1, n \rangle \}$

Derivation by Phase:

$\{ \{ [+connect] \}, \langle \emptyset_1, n \rangle \} \longrightarrow \{ \{ [+connect] \}, \langle \emptyset_1, n \rangle \}$ 【BSC】

The operable word-formation segment $\{\emptyset_1\}$ is transferred to the interface for operations (Phase S2 and S3 in Figure 1).

Structure Building: $\{ \{ [+connect] \}, nP \} \longrightarrow \{ \{ \emptyset \}, \{ \{ [+connect] \}, nP \} \}$

Merge-and-sequential-labeling stage:

$\{ \{ \emptyset \}, \{ \{ [+connect] \}, nP \} \} \longrightarrow \{ \{ \emptyset_2 \}, \{ \{ [+connect] \}, nP \} \}$

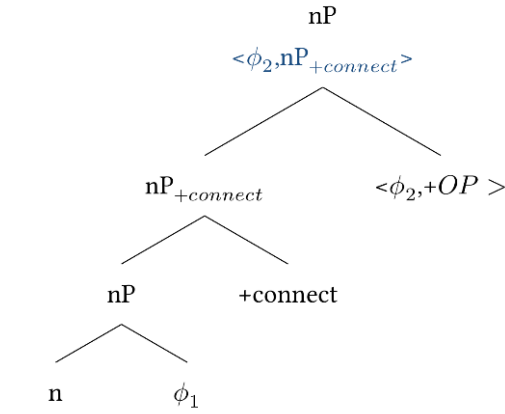
Derivation by Phase:

$\{ \{ \emptyset \}, \{ \{ [+connect] \}, nP \} \} \longrightarrow \{ \{ \emptyset_2 \}, \{ \{ [+connect] \}, nP \} \}$ 【BSC】

The word-formation segment derives to the top node and transfers the [+connect] segment to the interface for operations, completing the overall phrase derivation. See (4b).

(4b)

⁵ In the German word, the “s” is a linking consonant.



$\emptyset_1 \longleftrightarrow /gə'bu:r̥t/ \longrightarrow \sqrt{\text{Tag}}$

$[+connect] \longleftrightarrow /s/ \longrightarrow s$

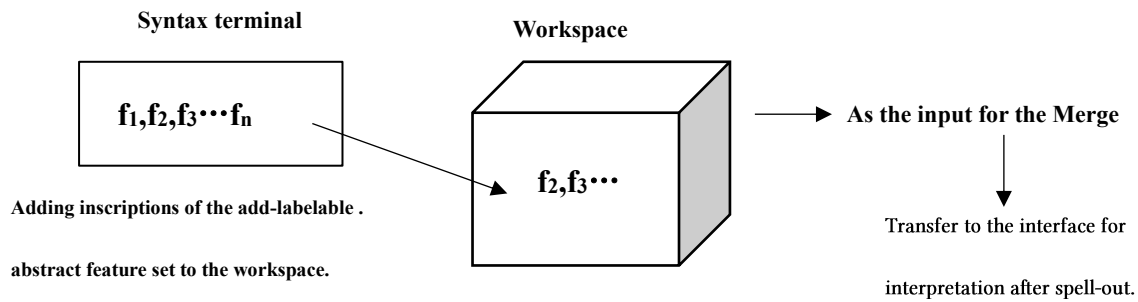
$\emptyset_2 \longleftrightarrow /_t ta:k/ \longrightarrow \sqrt{\text{Geburt}}$

$nP \longrightarrow \text{Geburtstag}_{[+n]}$

We can summarize as follows: In the syntactic derivation stage, the root occupies an empty position, which becomes an index at the spell-out stage. At the interface, phonology is realized and interpretation is provided through encyclopedic knowledge, all starting from the root as the baseline. Therefore, ROOT is a staged unit that unifies as both an index unit and a baseline for computation.

4. Implications of Root Duality for the Mechanisms of Syntactic Building Grammar under the New Constructionism

From the study of root duality, we have discovered that the methodology employed in lexical construction is essentially the phase derivation in syntactic construction. Consequently, it prompts us to reflect on what syntactic construction fundamentally operates upon. As Heidi Harley (2021) posits, the primitive elements of syntactic construction are the subclass abstract feature sets. Therefore, we contend that syntactic construction operates on these subclass abstract feature sets. When lexical construction is extended to the syntactic level, it essentially involves the inscription of labelable abstract feature sets being transferred to the workspace for merge (or agree operation) and subsequently delivered to the interface for interpretation, as illustrated in Figure 2.



**Figure 2. Language merging construction mechanism
under Neo-constructionism**

This paragraph outlines a Building-based view of language emergence, contrasting it with a generative vocabulary-centric view. It argues that humans are born with a basic language faculty and gradually acquire specific languages through maturation. In analysis of word formation and sentence construction, it suggests that the relevant syntactic terminal features are inscribed to form structures, which are then completed via the lexicon and encyclopedic knowledge. The root is not stored at the syntactic terminal; instead, it arises from an operable empty position created by merging, which is indexed at spell-out and used as the basis for construction at the interface.

5.Conclusion

We advocate that roots have different functions at different construction stages, and we argue that the syntactic terminal inventory in distributed morphology should not include roots. We also distinguish between generation and building, and contend that “building” is a more suitable term than “generation” when discussing word formation and sentence construction.

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